

Exploration of Transformer Ensemble and Auto Regressive Approaches to Enhance Performance of Clickbait Title Detection

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Abstract—“ClickBait” is a term which refers to online content titles designed to attract the viewers to click, often with the intention of misleading or exaggerating the actual content. Clickbait can be used for various malicious purposes such as social engineering, phishing, misinformation spreading, click fraud etc. Therefore, it is important to design a mechanism for detecting clickbait titles. For this purpose, we have utilised an auto regressive model called XLNet-base-case and 3 different types of transformer embedding models such as BERT-base-en-case, ROBERTa-base-en-case and XLM-ROBERTa-multi-case. We individually trained these models using a large diversified dataset and also performed ensemble by majority voting of BERT-base-en-case, ROBERTa-base-en-case and XLM-ROBERTa-multi-case eventually achieving test accuracy of 90% on imbalanced class dataset and 89% on balanced class dataset where the auto regressive approach gave validation accuracy of about 90.7% and 88.19% respectively on imbalanced class and balanced class datasets.

Index Terms—Clickbait, Transformer, BERT, ROBERTa, XLNet

I. INTRODUCTION

In the digital world, the spreading of online content has led to an increase in deceptive practices, notably through clickbait titles. These attractive headlines often mislead readers and contribute to misinformation. Clickbait titles weaken the integrity of online information, hamper the overall quality of content consumption and decrease the degree of user trustworthiness.

As a result, we need a robust and effective method for detecting clickbait titles. Recently, machine learning methods have been used to solve many of the problems in digital space. Therefore, the motivation behind this study is to apply ensemble [13] by majority voting of 3 transformer [15] based embedding models which are respectively BERT-base-en [4], ROBERTa-base-en-case [8] and XLM-ROBERTa-multi-case [3] and later on to apply auto regressive XLNet model [17] for enhancing the performance of clickbait title detection in order to erase the negative impacts such as social engineering, phishing, misinformation spreading, click fraud etc.

In this paper, we have developed and evaluated a transformer ensemble based system and compared with auto regressive

approach aiming to enhance the performance of clickbait title detection. In terms of transformer based models, after training all 3 of them individually, majority voting was applied on a hidden test dataset that was initially subtracted from the main large dataset. In terms of auto regressive approach, the model XLNet was tested on validation dataset after training.

We have observed and evaluated the difference between balanced and imbalanced class dataset prediction by leveraging transformer ensemble and auto regressive based approaches. We tried to generalize prediction for both clickbait and non-clickbait classes. We also tried to increase the diversity of the data and reduce overfitting in terms of clickbait title detection. For this purpose, we have collected a large dataset consisting of youtube [18] and news [7] contents.

Finally, we believe that our work will promote and ensure ethical content sharing by fighting against clickbait contents.

II. RELATED WORK

XAI is a field of methods and techniques that focuses on interpreting complex machine learning (black box) models. This section highlights literature review related to this work.

Understanding of XAI is inherited from this review paper[] which focuses on providing comprehensive analysis of XAI that comprises of current state of the art methods, taxonomy, fundamental concepts, challenges, applications and limitations of this concept. As LIME (an XAI technique) is mentioned as a human friendly explainer according the model agnostic methods comparison table 3, It is taken into account as one of the core contributors for calculating trust worthiness score of this novel research.

A. Early Works

A random forest classifier [12] has been used to train on the first clickbait corpus of 2992 Twitter tweets where 767 of which are clickbait. In terms of performance, 0.79 ROC-AUC at 0.76 precision and 0.76 recall is received.

In [1], a corpus created using multiple social media

platforms which was utilized in deep learning. Word2Vec [10] has been used for embedding purpose before training on (CNN) Convolutional Neural Networks. The sources for the dataset were respectively Reddit, Facebook and Twitter. Accuracy of about 90% received after word2vec word embedding and training on CNN.

B. Recent Works

In order to detect clickbait on youtube videos, multiple state-of-the-art machine learning techniques using a variety of textual features have been used [6]. In terms of word embeddings, Word2Vec, BERT, and DistilBERT [14] applied on a dataset containing about 8219 frequency of data. Out of those embedding models, BERT gave about 95% of accuracy.

A work [5] has been done on clickbait headline detection in Indonesian news sites using (M-BERT) Multilingual Bidirectional Encoder Representations [11] from Transformers where 6632 records were used for training purpose. After evaluating with 5-fold cross validation, it gave an accuracy of 91.4%.

By scraping data from Bengali online news articles and manually annotating them, Bengali [9] clickbait detector has been created by training using embedding models like BERT, BanglaBERT [2] and M-BERT. Finally, the best outcome out of these has been chosen as the final result.

Lack of diversity and frequency in data noticed in the related work section which we tried to resolve in our work of clickbait title detection.

III. METHODOLOGY

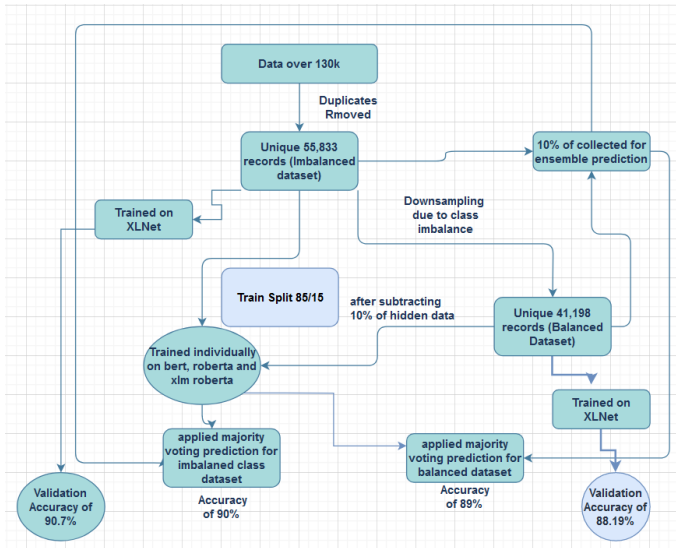


Fig. 1. Workflow of the Process

A. Dataset Information

For the purpose of clickbait text detection, we have collected a large dataset from Kaggle [16] [18] and Hugging Face [7]. The dataset contains records from youtube and news content. It is composed of 139,983 records of clickbait and non-clickbait text.

B. Data Preprocessing

From figure: 1 the initial collected data had some problems regarding duplicate records and class imbalance. First, we filtered the duplicate records. After filtering, we get a total unique records of 55,833. Out of these 55,833 records 20,599 turned out to be clickbait text and 35,234 records turned out to be non-clickbait text. As we can observe, there exist a class imbalance. So, from this point, we decided to follow two ways. One containing a dataset having balanced classes and the other one having imbalanced classes. We get the balanced dataset by undersampling the non-clickbait records.

For the imbalanced dataset, out of 55,833 records, we take 5,584 records as final hidden test for the purpose of ensemble prediction. After deducting, we have 50,249 records of data out of which 18,539 are clickbait contents and 31,710 are non-clickbait contents. For the balanced dataset, we randomly down sampled the non-clickbait unique records and got 41,198 frequency out of which 18,539 records are respectively clickbait and non-clickbait records.

C. Training and Testing Process

Figure: 1 also depicts the training and testing process. For the purpose of training, we used different embedding models such as BERT base case, ROBERTa base case and ROBERTa multilingual case following 2 layers which are respectively dropout layers used to help prevent overfitting and final dense layer for prediction having sig-moid activation function, Later on, we also used an auto regressive model called XLNet. During the process of train-test-split, we tried to make sure that the class proportion after the split remains the same on both train and test dataset.

First we individually trained for each of these models. We did some analysis for both balanced and imbalanced dataset. Then we performed majority voting of BERT-base-case, ROBERTa-base-case and ROBERTa-multilingual-case and compared the final test result with validation accuracy of XLNet model. For training purpose, we took 20 epochs for BERT-base-case, ROBERTa-base-case and ROBERTa-multilingual-case and around 3 epochs for XLNet-base-case. 85% of data considered for training and 15% of data considered for testing the individual word embedding models.

Finally, we performed prediction on the hidden test dataset by majority voting for both balanced and imbalanced dataset. Later on, we did analysis of the majority voting results with the XLNet model validation accuracy results.

IV. RESULTS AND DISCUSSION

A. Results Based on Imbalanced Class Dataset

Model	Validation Accuracy
BERT-base-12-en-cased	88%
ROBERTa-base-12-en-cased	90%
XLNet-base-cased	90.7%

TABLE I
RESULTS FOR IMBALANCED CLASS DATASET

In table I, we can observe that for imbalanced class dataset XLNet-base-cased model provides the best validation accuracy result of 90.7% which turns out to be slightly better than ROBERTa-base-12-en-cased word embedding model.

B. Results Based on Balanced Class Dataset

Model	Validation Accuracy
BERT-base-12-en-cased	87%
ROBERTa-base-12-en-cased	88%
XLNet-base-cased	88.19%

TABLE II
RESULTS FOR BALANCED CLASS DATASET

Similar pattern is noticed in table II when comparing the validation accuracy result on balanced class dataset. XLNet-base-cased model providing the highest validation accuracy of 88.19% which is slightly higher than ROBERTa-base-12-en-cased model. The rest of the models providing results less than 88%.

C. Majority Voting Results for both Balanced and Imbalanced Class Dataset

Class Dataset	Hidden Test Accuracy	Macro F1 Score
Imbalanced (55,833)	90%	89%
Balanced (41,198)	89%	89%

TABLE III
RESULTS FOR MAJORITY VOTING ON BERT, ROBERTA AND
XLNet-ROBERTA-MULTILINGUAL

In table III, we can observe the majority voting test set results for both imbalanced and balanced class dataset respectively providing test accuracy of 90% and 89%. For both cases, we observe same Macro F1 Score of 89% which means that they are achieving a similar overall balance between precision and recall across the two classes in their predictions.

D. Discussion According to The Results

After analyzing all of the results table, we can come up with several assumptions. For the case of imbalanced dataset, even after applying majority voting, we did not see any improvement in terms of test accuracy. Here, the majority voting test results for this case is the same as ROBERTa-cased-en validation accuracy. Interestingly, individual XLNet-base-case model surpasses this majority

voting result by 0.7% achieving the validation accuracy of 90.7%.

For the case of balanced dataset, we notice some improvements regarding hidden test accuracy. Here the majority voting performed better than all of the individual models validation accuracy, providing test accuracy of 89% where the best validation accuracy is 88.17% of XLNet-base-case which is slightly better than ROBERTa-en-cased. XLNet model can be a better choice for imbalanced class dataset where majority voting performs state of the art in terms of balanced class dataset on clickbait title detection.

Finally in this case, we can conclude that increasing the balanced dataset frequency can provide better performance if we apply majority voting on all of the models mentioned earlier.

V. LIMITATIONS

Although we went for several approaches regarding clickbait title detection, we faced some challenges during the process of accomplishing tasks. Due to lack of computational resources, we could not integrate XLNet-base-case autoregressive model as part of the ensemble process which could have led to a state of the art result as individually XLNet-base-case provided better results for both balanced and imbalanced class datasets.

Another limitation can be observed in the dataset class frequency. From the information provided in data preprocessing section, we notice a lack of clickbait text data due to which we might be getting results less enhanced than expected.

VI. CONCLUSION AND FUTURE WORK

In this paper, we explored the effectiveness of transformer ensemble based and auto-regressive approaches in improving the performance of clickbait title detection. Our findings reveal that performing the majority voting of respectively BERT-base-case-en, ROBERTa-base-case-en and XLNet-ROBERTa-multilingual-base embedding models provide better performance when dealing with balanced class dataset but auto-regressive approach performs state of the art when dealing with imbalanced class dataset.

In future, we plan to inject auto-regressive based model XLNet as part of the ensemble process which has a higher probability of providing better result than the other ones that also has been explored by us in this research work.

We can conclude that XLNet-base-case is the state of the art individual word embedding model in this regard and by increasing the balance and frequency of the dataset, utilizing majority voting of all the individual models can provide state of the art result which can be a future prospect.

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